# ANURAG KUMAR

## Vice President

Pune, India · +91 9356007435 · Kr1anurag@gmail.com · linkedin.com/in/anurag-kumar483a28212

## PROFESSIONAL PROFILE

Results-driven Technical Architect with 17+ years delivering scalable, fault-tolerant distributed systems in Java, Spring Boot, and cloud-native microservices across fintech, logistics, energy, and telecom. Expert in API-driven architecture, event-driven systems (Kafka), and polyglot persistence (MongoDB, Cassandra, Redis, Oracle). Proven track record in designing high-throughput distributed platforms, leading large engineering teams, and accelerating delivery through modern AI developer tooling.

## CORE COMPETENCIES

| | |
|---|---|
| **Languages & Frameworks** | Java 17 · Spring Boot 3+ · Python · PySpark · Angular 14 · TypeScript |
| **Architecture** | Microservices · API-Driven · Event-Driven · Cloud-Native · Distributed Systems |
| **AI Dev Tools** | GitHub Copilot · Devin AI |
| **Messaging & Cache** | Apache Kafka · Kafka Connect · Redis Cache |
| **Databases** | MongoDB · Cassandra · Oracle · SQL Server |
| **Cloud & DevOps** | Azure · AWS · GCP · Kubernetes · Docker · Prometheus · Grafana · CI/CD |
| **Big Data** | Azure Databricks · Azure Data Factory · PySpark · Synapse · ADLS Gen2 |

## WORK EXPERIENCE

### Vice President · Citi

*Pune | Oct 2024 – Present*

#### Application Design & Development

- Leading the modernisation of the legacy Actimize workflow to cloud-native microservices (Java 17, Spring Boot 3+, OpenShift) — 60% of components evaluated and migrated to the new microservice design, with the remaining modules in active development.
- Designed and implemented a RESTful API layer using Spring Boot 3+ to expose compliance screening, RFI management, and case resolution capabilities to downstream consumers, enforcing contract-first API design using OpenAPI specifications.
- Built an event-driven screening pipeline using Apache Kafka, decoupling compliance screening, case management, and SWIFT messaging systems for asynchronous, fault-tolerant communication at high throughput.
- Architected RFI automation platform on Spring Boot microservices with MongoDB (document store) and Oracle (transactional data), designing the service decomposition, data ownership boundaries, and API contracts across teams.
- Deployed all services on Kubernetes with a full observability stack — Prometheus for metrics, Grafana for dashboards, Loki for log aggregation, and Splunk for compliance audit trails.
- Defined and enforced microservice architecture standards across the team, covering service boundaries, API versioning, data consistency patterns, and deployment pipelines via CI/CD.
- Leveraged GitHub Copilot and Devin AI across the engineering team to accelerate development, code review, and test generation, reducing overall delivery time by up to 60%.

**Tech Stack:** *Java 17, Spring Boot 3+, Kafka, REST APIs, OpenAPI, Redis, MongoDB, Oracle, Kubernetes, Docker, Prometheus, Grafana, Loki, Splunk, CI/CD, GitHub Copilot, Devin AI*

## Senior Technical Architect · Maersk

*Bangalore | Oct 2022 – Oct 2024*

### Application Design & Development

- Modernised legacy Delphi-based logistics systems to a cloud-native microservices stack on Azure, improving scalability, reliability, and maintainability for a high-traffic global platform.
- Built inland transportation platform as independent Spring Boot microservices covering freight booking, dispatch, multi-modal vendor management (road, rail, ocean, air), tracking, and last-mile delivery — integrated via Kafka and REST APIs.
- Designed event-driven Kafka backbone for async, decoupled communication across booking, dispatch, tracking, and billing services.
- Adopted polyglot persistence: MongoDB for freight documents, Cassandra for high-write tracking events, SQL Server for transactional billing.
- Built Angular 14 dispatch planning and vendor management dashboards used by global operations teams.
- Delivered Azure Databricks and Azure Data Factory pipelines for operational analytics and transportation planning dashboards.
- Led and mentored a 30+ member engineering team; drove architecture standards and Agile delivery.

### Agentic AI · Landside Transportation Automation

- Architected a stateful Multi-Agent LangGraph workflow for landside freight order validation — agents autonomously check carrier availability, route feasibility, customs compliance, and tariff eligibility before confirming dispatch.
- Built a RAG pipeline over carrier contracts, tariff schedules, and customs regulation documents using Azure OpenAI embeddings stored in Azure Cognitive Search, enabling agents to make grounded routing and vendor selection decisions in real time.
- Designed agent state machine with persistent checkpointing in MongoDB, enabling long-running freight workflows to survive system restarts and resume from the last verified state.
- Added human-in-the-loop approval gates for high-risk exceptions (hazmat, oversize cargo, cross-border), with agents autonomously resuming execution and updating downstream booking and billing services post-approval.
- Registered internal logistics services (carrier availability API, shipment tracking, ETA prediction, customs clearance status) as MCP servers, enabling agents to query and invoke these tools dynamically based on the freight order context at hand.
- Monitored agent reasoning quality via a custom observability dashboard (LangSmith + Grafana) tracking tool call latency, RAG retrieval scores, and decision confidence per freight order.

**Tech Stack:** *Java 17, Spring Boot, Python, LangChain, LangGraph, MCP, RAG, Azure OpenAI, Azure Cognitive Search, Kafka, Redis, MongoDB, Cassandra, SQL Server, Angular 14, TypeScript, PySpark, Kubernetes, Azure Databricks, ADF, Synapse, LangSmith, Grafana*

## AVP · Technical Architect · Credit Suisse

*Pune | Jan 2021 – Oct 2022*

- Built and enhanced Product Buffet — a Wealth Management platform providing tactical overview of Fund Profile, Risk, Performance, Returns, Disclaimers, and SMR as a product one-pager — using Spring Boot microservices backend and Angular frontend.
- Part of the key product strategy team for various applications across the Wealth Management portfolio; contributed to feature roadmap, API design, and architecture governance.
- Defined current-state architecture, mapped future-state architecture, and designed the transition roadmap, aligning technical delivery with business goals.
- Modernised deployment strategy from on-premise datacenters to OpenShift and Azure Cloud, improving scalability, resilience, and release velocity.
- Led design and development of multiple modules; implemented architecture principles, strategies, and guidelines across development teams.
- Collaborated with stakeholders to understand functional and non-functional requirements and translate them into technical solutions.
- Applied ML-based models for fund risk scoring and performance classification — trained classification algorithms (Random Forest, Gradient Boosting) on historical NAV and volatility data to categorise fund risk levels with measurable accuracy improvements over rule-based approaches.

- Evaluated model quality using standard metrics including accuracy, precision, recall, and F1-score; iterated on feature engineering and hyperparameter tuning to improve prediction reliability for risk and return profiles.
- Leveraged Azure Databricks and PySpark for model training pipelines over large fund datasets stored in ADLS Gen2; used Azure ML for experiment tracking and model versioning.

**Tech Stack:** *Java 11, Spring Boot, Python, MongoDB, Cassandra, Kubernetes, Azure, Angular, Azure Databricks, PySpark, ADF, ADLS Gen2, Azure ML, Scikit-learn*

### Technical Lead · Schlumberger

*Pune | Feb 2020 – Dec 2020*

- Led the GAIA project ingesting and cleansing high-volume geographical and IoT sensor data from Oil & Gas sites using PySpark on GCP, forming the data backbone for geoscientist research workloads.
- Built Spring Boot microservices and Angular 9 modules for full-stack delivery; mentored junior developers and maintained code quality under tight timelines.

**Tech Stack:** *Java 8, Python, PySpark, Spring Boot, Kafka, Angular 9, GCP*

### Senior Manager · Vodafone

*Pune | Jul 2017 – Feb 2020*

- As part of the Global Integration Gateway (GIG) programme, led the end-to-end redesign of Pay-to-Bill and Partner Provisioning services, encompassing OSS, BSS, SecureNet, and VPN integration layers.
- Led project planning, design, execution, and delivery to replace the legacy TIBCO ecosystem with an event-driven microservices architecture on AWS and Kubernetes using Kafka and Redis.
- Collaborated with Product Owners, IT Architects, and Business stakeholders to identify functional and non-functional requirements and define end-to-end solution architecture.
- Partnered with the platform team to drive digital transformation on AWS cloud and Kubernetes, establishing CI/CD pipelines and cloud-native deployment standards.
- Responsible for designing monitoring and observability solutions using Prometheus and Grafana for real-time platform health visibility.
- Expanded the engineering team from 3 to 50 members to develop new products including the GIG platform and Chatbot services.
- Drove product delivery, stakeholder management, and people management across a distributed team.

**Tech Stack:** *AWS, Spring Boot, Kafka, Redis, Docker, Kubernetes, MongoDB, CI/CD, Jenkins, Prometheus, Grafana, AWS Glue, Athena*

### Technical Lead · Sears

*Pune | Sep 2015 – May 2017*

- Module lead for the Financial Transaction and Settlement division — owned end-to-end microservices for high-value bank settlement across Sears and Kmart stores.
- Redesigned settlement pipeline with Kafka Connect and Kafka Streams for real-time processing; built PySpark batch jobs replacing costly legacy ETL, delivering high throughput at significantly reduced infrastructure cost.

**Tech Stack:** *Java 8, Spring Boot, Kafka, Kafka Connect, PySpark, AWS, Angular 1.4, Control-M, SQL Server*

### Senior Software Developer · Talentica

*Pune | Aug 2012 – Sep 2015*

- Full-stack development of a life insurance platform with underwriter, e-App, and PA capabilities.

**Tech Stack:** *Java 8, Angular 2, AWS, Kafka, Redis, Cassandra, SQL Server, ElasticSearch*

### Senior Software Developer · CSC

*Bangalore | Apr 2008 – Jul 2012*

- Development and support of a life insurance platform with Underwriter capabilities.

**Tech Stack:** *Java 5, JSF, Hibernate, XML, SOAP*

## KEY ACCOMPLISHMENTS

| | |
|---|---|
| **Agentic AI – Landside (Maersk)** | Stateful Multi-Agent LangGraph workflow for autonomous freight validation, vendor selection, and dispatch confirmation. |
| **RAG – Compliance Docs (Maersk)** | RAG pipeline over carrier contracts, tariff schedules, and customs regulations using Azure OpenAI + Azure Cognitive Search for real-time grounded routing decisions. |
| **Event-Driven Architecture** | Kafka-based platforms across Maersk, Vodafone, and Sears for resilient, high-throughput processing. |
| **Cost-Efficient Delivery (Sears)** | Kafka and PySpark pipelines replaced legacy ETL delivering high throughput at significantly lower infrastructure cost. |
| **AI Developer Tooling** | GitHub Copilot and Devin AI adopted across Citi team, reducing delivery time by up to 60%. |
| **Team Leadership** | Built and led teams of 30 to 50 engineers across fintech, logistics, energy, and telecom programs. |

## EDUCATION & CERTIFICATIONS

| | |
|---|---|
| **B.Tech - Electronics & Telecommunication**<br>National Institute of Technology (NIT), Silchar · 2007 | **CKAD - Certified Kubernetes Application Developer**<br>February 2020<br>**AWS Certified Solutions Architect**<br>March 2021 |